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Linear Algebra

Fall 2026

This document presents selected detailed solutions of Week 6 Practical Session exercises

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The document contains:

1. Detailed solution of problem 20 (from Week 6 - Practical Session exercises).
2. Detailed solutions of problems 9-12 (from Week 6 - Practical Session exercises).
3. Detailed solutions of problem 17 (from Week 6 - Practical Session exercises).

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1. Detailed solution of problem 20

(Challenge problem from Week 6 - Practical Session exercises)

Problem 20 – Challenge: tridiagonal model

Solve $Ax = b$ by LU for $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$. Interpret the nearest-neighbor structure.

Solution of problem 20

We need to solve

$$A\mathbf{x} = \mathbf{b}, \qquad A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix},$$

using **LU factorization**, and interpret the **nearest-neighbor structure**.

1. Factor $A = LU$

Eliminate the entry -1 below the first pivot 2. The multiplier is

$$m_{21} = \frac{-1}{2} = -\frac{1}{2}.$$

Therefore,

$$R_2 \leftarrow R_2 - m_{21} R_1 = R_2 + \frac{1}{2} R_1,$$

giving

$$\begin{pmatrix} 2 & -1 & 0 \\ 0 & \frac{3}{2} & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

Next, eliminate the entry below the second pivot:

$$m_{32} = \frac{-1}{3/2} = -\frac{2}{3}, \qquad R_3 \leftarrow R_3 + \frac{2}{3}R_2.$$

The final diagonal entry becomes $2 - \frac{2}{3} = \frac{4}{3}$. Thus,

$$\boxed{L = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 2 & -1 & 0 \\ 0 & \frac{3}{2} & -1 \\ 0 & 0 & \frac{4}{3} \end{pmatrix}.}$$

The elimination multipliers appear directly in L, including their negative signs.

2. Forward substitution: solve $L\mathbf{y} = \mathbf{b}$

$$\begin{cases} y_1 = 3, \\ -\frac{1}{2}y_1 + y_2 = 0, \\ -\frac{2}{3}y_2 + y_3 = 2. \end{cases}$$

Consequently,

$$y_1 = 3, \qquad y_2 = \frac{3}{2}, \qquad y_3 = 2 + \frac{2}{3}\left(\frac{3}{2}\right) = 3.$$

Hence

$$\mathbf{y} = \begin{pmatrix} 3 \\ \frac{3}{2} \\ 3 \end{pmatrix}.$$

3. Back substitution: solve $U\mathbf{x} = \mathbf{y}$

$$\begin{cases} 2x_1 - x_2 = 3, \\ \frac{3}{2}x_2 - x_3 = \frac{3}{2}, \\ \frac{4}{3}x_3 = 3. \end{cases}$$

Starting from the last equation,

$$x_3 = \frac{9}{4},$$

$$x_2 = \frac{2}{3}\left(\frac{3}{2} + \frac{9}{4}\right) = \frac{5}{2},$$

$$x_1 = \frac{3 + x_2}{2} = \frac{11}{4}.$$

Therefore,

$$\boxed{\mathbf{x} = \begin{pmatrix} \frac{11}{4} \\ \frac{5}{2} \\ \frac{9}{4} \end{pmatrix}.}$$

Checking in the original equations:

$$2\left(\tfrac{11}{4}\right) - \tfrac{5}{2} = 3, \qquad -\tfrac{11}{4} + 2\left(\tfrac{5}{2}\right) - \tfrac{9}{4} = 0, \qquad -\tfrac{5}{2} + 2\left(\tfrac{9}{4}\right) = 2.$$

4. Interpret the nearest-neighbor structure

The equations are

$$\begin{aligned} 2x_1 - x_2 &= 3, \\ -x_1 + 2x_2 - x_3 &= 0, \\ -x_2 + 2x_3 &= 2. \end{aligned}$$

The matrix is **tridiagonal**: nonzero entries occur only on the main diagonal and immediately above and below it.

- x_1 interacts directly with x_2.
- x_2 interacts directly with both x_1 and x_3.
- x_3 interacts directly with x_2.
- There is **no direct coupling between** x_1 **and** x_3, because $A_{13} = A_{31} = 0$.

This structure occurs in models of **heat conduction along a rod, chains of springs, and one-dimensional finite-difference equations**. In particular,

$$-x_{i-1} + 2x_i - x_{i+1}$$

is the standard discrete negative second derivative, apart from the grid-spacing factor. The specific physical interpretation depends on the model and its boundary conditions.

Notice that elimination preserves the narrow structure: L **and** U **are bidiagonal**, making this type of system inexpensive to solve.

5. LDU and normalized U for this matrix

Extract the pivots from U:

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & \frac{4}{3} \end{pmatrix}.$$

Divide each row of U by its pivot:

$$\widehat{U} = D^{-1}U = \begin{pmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & 1 & -\frac{2}{3} \\ 0 & 0 & 1 \end{pmatrix}.$$

Thus,

$$\boxed{A = LD\widehat{U}.}$$

2. Detailed solutions of problems 9-12

(from Week 6 - Practical Session exercises)

We use **Gaussian elimination** to obtain an upper triangular system, then solve by **back substitution**. Problem #10 has infinitely many solutions; the other three have unique solutions.

Throughout, write $\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$.

Problem 9

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 2 \\ 6 \\ 1 \end{pmatrix}.$$

The system is

$$\begin{cases} x + y + z = 2, \\ 2x + 3y + 2z = 6, \\ x + 2y + 4z = 1. \end{cases}$$

Start with its augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 4 & 1 \end{array}\right].$$

Eliminate x from rows 2 and 3:

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - R_1.$$

For example, the new second row is

$$(2, 3, 2 \mid 6) - 2(1, 1, 1 \mid 2) = (0, 1, 0 \mid 2).$$

Thus,

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 3 & -1 \end{array}\right].$$

Eliminate y from row 3:

$$R_3 \leftarrow R_3 - R_2,$$

giving

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 3 & -3 \end{array}\right].$$

The resulting triangular system is

$$\begin{cases} x + y + z = 2, \\ y = 2, \\ 3z = -3. \end{cases}$$

Back substitution gives

$$z = -1, \qquad y = 2, \qquad x = 2 - y - z = 2 - 2 + 1 = 1.$$

Therefore,

$$\boxed{\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}.}$$

Check:

$$A\mathbf{x} = \begin{pmatrix} 1 + 2 - 1 \\ 2 + 6 - 2 \\ 1 + 4 - 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \\ 1 \end{pmatrix} = \mathbf{b}.$$

Problem 10

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 5 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 4 \\ 8 \\ 2 \end{pmatrix}.$$

The system is

$$\begin{cases} x + 2y + z = 4, \\ 2x + 4y + 2z = 8, \\ x + 3y + 5z = 2. \end{cases}$$

Its augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 1 & 3 & 5 & 2 \end{array}\right].$$

Eliminate x below the first pivot:

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - R_1.$$

We obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 4 & -2 \end{array}\right].$$

The zero row represents $0 = 0$, which imposes **no additional restriction**. Indeed, the original second equation is exactly twice the first.

Exchange rows 2 and 3 to put the matrix in row echelon form:

$$R_2 \leftrightarrow R_3.$$

Then

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 4 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

There are two pivots and three unknowns, so one variable is free. Choose

$$z = t, \qquad t \in \mathbb{R}.$$

The second row gives

$$y + 4z = -2 \quad \Longrightarrow \quad y = -2 - 4t.$$

The first row gives

$$x = 4 - 2y - z = 4 - 2(-2 - 4t) - t = 8 + 7t.$$

Therefore, **all solutions** are

$$\boxed{x = 8 + 7t, \qquad y = -2 - 4t, \qquad z = t, \quad t \in \mathbb{R}.}$$

Equivalently,

$$\boxed{\mathbf{x} = \begin{pmatrix} 8 \\ -2 \\ 0 \end{pmatrix} + t \begin{pmatrix} 7 \\ -4 \\ 1 \end{pmatrix}, \qquad t \in \mathbb{R}.}$$

Check the first and third equations:

$$(8 + 7t) + 2(-2 - 4t) + t = 4,$$

$$(8 + 7t) + 3(-2 - 4t) + 5t = 2.$$

The second equation also holds because it is twice the first.

Geometrically: the first two equations describe the same plane. Its intersection with the third plane is a line—the solution family above.

Problem 11

$$A = \begin{pmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 6 \\ 10 \\ 5 \end{pmatrix}.$$

The system is

$$\begin{cases} x + 3y + z = 6, \\ 2x + 5y + 2z = 10, \\ x + 4y + 4z = 5. \end{cases}$$

Write

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 4 & 5 \end{array}\right].$$

Eliminate x below the first pivot:

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - R_1.$$

This produces

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 6 \\ 0 & -1 & 0 & -2 \\ 0 & 1 & 3 & -1 \end{array}\right].$$

The second pivot is -1. To eliminate the 1 beneath it, **add** row 2 to row 3:

$$R_3 \leftarrow R_3 + R_2.$$

Hence,

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 6 \\ 0 & -1 & 0 & -2 \\ 0 & 0 & 3 & -3 \end{array}\right].$$

Back substitution:

$$3z = -3 \quad \Longrightarrow \quad z = -1,$$

$$-y = -2 \quad \Longrightarrow \quad y = 2,$$

$$x = 6 - 3y - z = 6 - 6 + 1 = 1.$$

Therefore,

$$\boxed{\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}.}$$

Check:

$$A\mathbf{x} = \begin{pmatrix} 1+6-1 \\ 2+10-2 \\ 1+8-4 \end{pmatrix} = \begin{pmatrix} 6 \\ 10 \\ 5 \end{pmatrix} = \mathbf{b}.$$

Problem 12

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 2 \\ 6 \\ 2 \end{pmatrix}.$$

The system is

$$\begin{cases} x + y + z = 2, \\ 2x + 3y + 2z = 6, \\ x + 2y + 3z = 2. \end{cases}$$

Start with

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 3 & 2 \end{array}\right].$$

Eliminate x below the first pivot:

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - R_1.$$

We obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 2 & 0 \end{array}\right].$$

Next,

$$R_3 \leftarrow R_3 - R_2,$$

so

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 2 & -2 \end{array}\right].$$

Back substitution gives

$$2z = -2 \quad \Longrightarrow \quad z = -1,$$

$$y = 2,$$

$$x = 2 - y - z = 1.$$

Therefore,

$$\boxed{\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}.}$$

Check:

$$A\mathbf{x} = \begin{pmatrix} 1+2-1 \\ 2+6-2 \\ 1+4-3 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \\ 2 \end{pmatrix} = \mathbf{b}.$$

Comparison of the four systems

Problem	Pivots after elimination	Conclusion
9	$1, 1, 3$	Unique solution: $(1, 2, -1)$
10	Two pivots; one zero row	Infinitely many solutions: $(8 + 7t, -2 - 4t, t)$
11	$1, -1, 3$	Unique solution: $(1, 2, -1)$
12	$1, 1, 2$	Unique solution: $(1, 2, -1)$

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3. Detailed solution of problem 17

(from Week 6 - Practical Session exercises)

Find a row permutation P and factors satisfying $PA = LU$ for

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 2 \end{bmatrix}.$$

Solution:

We want to find a **permutation matrix** P, a **unit lower triangular matrix** L, and an **upper triangular matrix** U such that

$$\boxed{PA = LU}, \qquad A = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 2 \end{pmatrix}.$$

1. Exchange rows to obtain a nonzero first pivot

The first diagonal entry is 0, so we cannot use it as a pivot. Exchange rows 1 and 2:

$$R_1 \leftrightarrow R_2.$$

The corresponding permutation matrix is obtained by making the same exchange in the identity matrix:

$$\boxed{P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.}$$

Multiplying A by P on the **left** exchanges its rows:

$$PA = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{pmatrix}.$$

We now factor this permuted matrix.

2. Eliminate below the first pivot

The first pivot is 1. The entry in position $(2, 1)$ is already zero, so

$$m_{21} = 0.$$

To eliminate the 2 in position $(3, 1)$, use the multiplier

$$m_{31} = \frac{2}{1} = 2.$$

Apply

$$R_3 \leftarrow R_3 - 2R_1.$$

The third row becomes

$$(2, 3, 2) - 2(1, 1, 0) = (0, 1, 2).$$

Thus,

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

3. Eliminate below the second pivot

The second pivot is 2. The multiplier is

$$m_{32} = \tfrac{1}{2}.$$

Apply

$$R_3 \leftarrow R_3 - \tfrac{1}{2}R_2.$$

The third row becomes

$$(0, 1, 2) - \tfrac{1}{2}(0, 2, 1) = \left(0, 0, \tfrac{3}{2}\right).$$

Therefore,

$$\boxed{U = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & \frac{3}{2} \end{pmatrix}.}$$

4. Construct L from the elimination multipliers

Place ones on the diagonal and put each multiplier in its corresponding position below the diagonal:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{pmatrix}.$$

Hence,

$$\boxed{L = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & \frac{1}{2} & 1 \end{pmatrix}.}$$

Notice that L contains the multipliers 2 **and** $\tfrac{1}{2}$, even though the row operations subtract these multiples.

5. Verify $PA = LU$

Multiply the factors:

$$LU = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & \frac{3}{2} \end{pmatrix}.$$

The first two rows reproduce the first two rows of U. The third row is

$$2(1, 1, 0) + \tfrac{1}{2}(0, 2, 1) + \left(0, 0, \tfrac{3}{2}\right) = (2, 3, 2).$$

Thus,

$$LU = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{pmatrix} = PA.$$

The required factorization is therefore

$$\underbrace{\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{P} \underbrace{\begin{pmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 2 \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & \frac{1}{2} & 1 \end{pmatrix}}_{L} \underbrace{\begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & \frac{3}{2} \end{pmatrix}}_{U}.$$

Interpretation: P first reorders the equations to provide a usable pivot. Gaussian elimination then

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